

Day 8 — First iFlow, HTTPS & Mail Adapter Usage (25 Aug 2026)

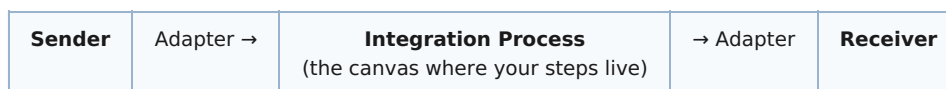
1. iFlow Creation — File/Flow Naming

iFlow creation — naming convention used in this session's example: **Day_08_https_mail_adapter**

2. Talking About Version — Draft Status

When you first create an iFlow, its version status is **Draft** — meaning it hasn't been deployed/activated yet, it's still being worked on.

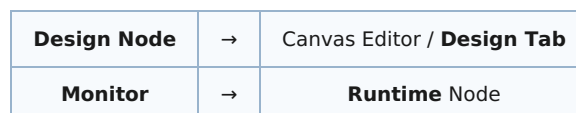
3. The Basic iFlow Structure (Diagram)



This is the fundamental skeleton of every iFlow: a Sender connects in via an Adapter, your logic runs inside the Integration Process, and the result goes out to a Receiver via another Adapter.

4. Two Types of Nodes

Node Type	Where It Lives / What It Means
Design Node	The Canvas editor — this is the Design Tab , where you build the iFlow
Runtime Node	Where the iFlow actually executes once deployed — tracked in the Monitor Tab



5. About Adapters — Pre-Built Backend

Adapters → already developed Backend. This is an important point: you as a developer don't build adapters from scratch — SAP has already built and maintains the backend logic for each adapter type; you just configure them.

6. CPI's Underlying Technology

CPI → Runs on Apache Camel framework

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Java programming language

Confirms earlier material: This lines up with what we covered back on Day 1's architecture discussion — Apache Camel is a java-based, open-source integration framework, and it's the engine underneath CPI's graphical tools.

7. Palette Functions Usage

Palette functions usage — the palette is the panel in the Design tab where all the available steps (Content Modifier, Router, Splitter, etc.) live, ready to be dragged onto the canvas.

8. Now: Basic Flow Creation (Hands-On)

HTTPS Adapter — Connection Tab Fields

Field	Purpose
Request Processing / Address	Defines the endpoint path for this sender
Authorization	Access control setting
User Role	Restricts which roles can call this endpoint
CSRF Protected	Whether CSRF protection is enabled for this endpoint

Sender to CPI — Creating the Address

Sender to CPI → we created Address. This is the unique path segment identifying this specific iFlow's endpoint.

Endpoint Creation — Where It Happens

Endpoint created	at	Runtime
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Trial → Design — meaning: in a trial tenant, this endpoint creation happens through the Design tab process (as opposed to some more complex enterprise CI/CD-driven activation pattern).

A unique name we will give to create the endpoint — each sender endpoint address must be unique within your tenant.

9. Service Key

Service Key explanation — covered as a concept in this session (tying back to the credentials/authentication topic from Day 7).

10. About Postman Tool

Sender → data → validate in POSTMAN whether the data is receiving correctly or not. Postman is used here as the external tool to actually trigger your iFlow's HTTPS endpoint and confirm it behaves as expected.

Testing Result

- **Sample iFlow created and tested from Postman**
- **200 OK Status Code** — confirms the request was successfully received and processed

After Success — Where to Check the Message

After success, seen in Monitor, where we need to see the message.

Important operational point: CPI doesn't store the data itself (a principle we covered back on Day 2) — it passes data through. So to actually inspect message content after the fact, you rely on **Log Configuration → Trace option**, since the raw message body isn't persisted by default.

11. Log Configuration — Trace Option

Overview → Messages → Log Level Options

Log Level	→	Trace
	→	Info

Trace level → captures **Message Content** (the actual payload), which Info level does not.

Important Rule: Starting Point & Production Caution

- **We need to leave/set a "Starting Point"** when enabling trace
- **Production level: We should NOT turn on "Trace" carelessly** — this is a genuine best practice, since Trace persists full message content (which can include sensitive data), and has real cost/performance/compliance implications in a live system

Worked Example — Message Transformation Visualized via Trace

Sender	→	CPI	→	Receiver System
"Hi @all, BTP is welcome to CPI class"		(Content Modifier → Message Body)		

Content Modifier sets Message Body to: "Hi everyone, welcome to CPI/BTP CPI IS class"

Here, "@all" is replaced by "everyone" — this is the Transformation in action, visible clearly when you inspect the message via Trace.

Trace Timing — Exact Numbers (Verified)

Verified accurate against official SAP documentation: Once Trace is enabled, it stays active for **10 minutes** (after which the log level automatically reverts to whatever it was set to before). The actual trace content/payload captured during that window remains available (retained) for approximately **1 hour (60 minutes)** before being removed. This matches exactly what your notes captured — good, precise detail to remember for real project work.

12. Mail Adapter Usage

The Mail adapter's configuration is organized across four tabs:

- **General**
- **Connection**
- **Processing**
- **Security**

Gmail Host & Port

Setting	Value
Host	smtp.gmail.com
Port	587

Verified accurate: Port 587 is Gmail's standard SMTP submission port using STARTTLS encryption — correct for sending mail via CPI's Mail adapter.

Connection Tab — Full Field List

Field	Example / Notes
Address	smtp.gmail.com:587
Proxy Type	Internet, Proxy
Timeout	30000 (milliseconds, i.e., 30 seconds)
Protection	(connection security setting)
Authentication Type	(method used to authenticate to the mail server)
Credential Name	Reference to your stored security material

Creating an App Password for Gmail

Creation of App Password — for Gmail account. Since Gmail requires an app-specific password (not your normal login password) for third-party SMTP access like this, this is a required setup step before CPI can authenticate.

Creating User Credentials in Security Material

Creation of User Credentials in Security Material — named as: **Gmail(or)gmail** (i.e., your trainer used a naming convention like "Gmail" or lowercase "gmail" as the Security Material credential name — exact casing depends on what you name it in your own tenant).

Processing Tab — Email Fields

Field	Purpose

From	Sender email address
To	Primary recipient(s)
CC	Carbon copy recipient(s)
BCC	Blind carbon copy recipient(s)
Subject	Email subject line
Mail Body	The actual email content/message